INCH-POUND
MIL-DTL-24640/17E
8 November 2011
SUPERSEDING
MIL-DTL-24640/17D
22 August 2002

DETAIL SPECIFICATION SHEET

# CABLE, ELECTRIC, -20 TO +105 °C, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR, 26 AWG, TYPE 2XOW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

## Construction (Watertight)

| | | |
|---|---|---|
| First | - | Tin-coated copper conductor in accordance with table I. |
| Second | - | Insulation material and basic construction requirements shall be in accordance with SAE-AS81044/12. Testing shall be in accordance with the component wire ratings and requirements listed on this specification sheet. Conformance testing shall be in accordance with SAE-AS81044 prior to cabling. |
| Third | - | Two conductors twisted together. Pair lay length shall be in accordance with table III. Twisted pair identification code, one black and one white, applied using Method 6, or telephone color code applied using Method 3. For 2XOW-77, if telephone color code is used, repeat the color code of pairs 1-11, in sequence, for pairs 67-77. |
| Fourth | - | The required number of twisted pairs shall be cabled together in accordance with table II with a lay length not greater than 16 times the diameter over the outermost layer. Cable sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used, as necessary, to form a round cable. |
| Fifth | - | Optional binder tape. |
| Sixth | - | Braided shield, tin-coated copper strands. Braid must be constructed to meet surface transfer impedance requirements as listed herein. |
| Seventh | - | Optional binder tape. |
| Eighth | - | Cross-linked polyolefin jacket (see table II for thickness).<br>NOTE: Manufacturer's identification tape or thread is required (see 3.7.3). |



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TABLE I. Component wire details.

| Watertight conductor | | Finished wire | | |
|---|---|---|---|---|
| Wire size (AWG) | Stranding (no. x AWG) | Test load life cycle accelerated aging tests (lbs ±3%) | Low temp. at -40 °C | |
| | | | Mandrel diameter (inches) | Test load (lbs) |
| 26 | 7 x 34 | 0.125 | 0.50 | 0.50 |

TABLE II. Construction details.

| PIN M24640/17 | Type number | Number of pairs | Jacket thickness (inch) (min.) | Cable diameter (inch) | | Cable weight (pounds per 1000 feet) (max.) |
|---|---|---|---|---|---|---|
| | | | | (min.) | (max.) | |
| -01UO | 2XOW-6 | 6 | 0.038 | 0.336 | 0.363 | 181.6 |
| -02UO | 2XOW-18 | 18 | 0.041 | 0.468 | 0.504 | 201.0 |
| -03UO | 2XOW-24 | 24 | 0.044 | 0.546 | 0.588 | 266.0 |
| -04UO | 2XOW-42 | 42 | 0.044 | 0.646 | 0.686 | 380.0 |
| -05UO | 2XOW-60 | 60 | 0.047 | 0.744 | 0.802 | 485.0 |
| -06UO | 2XOW-77 | 77 | 0.047 | 0.840 | 0.906 | 615.0 |



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TABLE III. Pair lay [1/].

| Pair no. | Pair lay (inches) | Pair no. | Pair lay (inches) | Pair no. | Pair lay (inches) |
|---|---|---|---|---|---|
| 1 | 1.5 | 27 | 1.0 | 53 | 2.5 |
| 2 | 1.0 | 28 | 2.0 | 54 | 3.0 |
| 3 | 2.0 | 29 | 3.0 | 55 | 4.5 |
| 4 | 3.0 | 30 | 4.5 | 56 | 5.5 |
| 5 | 4.5 | 31 | 1.0 | 57 | 1.0 |
| 6 | 1.0 | 32 | 2.0 | 58 | 1.5 |
| 7 | 2.0 | 33 | 3.0 | 59 | 2.0 |
| 8 | 3.0 | 34 | 4.5 | 60 | 2.5 |
| 9 | 1.5 | 35 | 1.0 | 61 | 3.0 |
| 10 | 2.5 | 36 | 2.0 | 62 | 4.5 |
| 11 | 5.5 | 37 | 3.0 | 63 | 5.5 |
| 12 | 1.5 | 38 | 4.5 | 64 | 1.0 |
| 13 | 2.5 | 39 | 1.0 | 65 | 1.5 |
| 14 | 3.7 | 40 | 2.0 | 66 | 2.0 |
| 15 | 5.5 | 41 | 3.0 | 67 | 2.5 |
| 16 | 1.5 | 42 | 4.5 | 68 | 3.0 |
| 17 | 2.5 | 43 | 1.0 | 69 | 4.5 |
| 18 | 3.7 | 44 | 1.5 | 70 | 5.5 |
| 19 | 1.5 | 45 | 2.0 | 71 | 1.0 |
| 20 | 2.5 | 46 | 2.5 | 72 | 1.5 |
| 21 | 3.7 | 47 | 3.0 | 73 | 2.0 |
| 22 | 5.5 | 48 | 4.5 | 74 | 2.5 |
| 23 | 1.0 | 49 | 5.5 | 75 | 3.0 |
| 24 | 2.0 | 50 | 1.0 | 76 | 4.5 |
| 25 | 3.0 | 51 | 1.5 | 77 | 5.5 |
| 26 | 4.5 | 52 | 2.0 | | |

NOTE:
[1/] All lays are left-handed.

TABLE IV. Assembly.

| Type number | Core | Assembly | | | |
|---|---|---|---|---|---|
| | | Layer 1 | Layer 2 | Layer 3 | Layer 4 |
| 2XOW-6 | --- | Pairs 1-6 | --- | --- | --- |
| 2XOW-18 | Pair 1 | Pairs 2-6 | Pairs 7-18 | --- | --- |
| 2XOW-24 | Pairs 1-2 | Pairs 3-10 | Pairs 11-24 | --- | --- |
| 2XOW-42 | --- | Pairs 1-8 | Pairs 9-22 | Pairs 23-42 | --- |
| 2XOW-60 | Pair 1 | Pairs 2-7 | Pairs 8-19 | Pairs 20-36 | Pairs 37-60 |
| 2XOW-77 | Pairs 1-3 | Pairs 4-12 | Pairs 13-27 | Pairs 28-49 | Pairs 50-77 |



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REQUIREMENTS:

Qualification required.

Construction details: See tables I, II, III, and IV.

Component wire: The component wire shall be in accordance with SAE-AS81044 and shall be tested in accordance with the following wire ratings and additional requirements.

| | |
|---|---|
| Temperature rating | 135 °C (275 °F) (max.) conductor temperature |
| Voltage rating | 600 volts (root mean square) at sea level |
| Accelerated aging | Oven temperature, 250±2 °C (482±3.6 °F) for 6 hours |
| Low temp. cold bend | -40 °C (-40 °F) for 4 hours |
| Blocking | 150±2 °C (302±3.6 °F) |
| Color | In accordance with MIL-STD-104, Class 1; white preferred |
| Flammability | 30 seconds (max.) |
| | 3.0 inches (max.) |
| | No flaming of tissue paper |
| Humidity resistance | 5000 megohms for 1000 feet (min.) insulation resistance after humidity exposure |
| Dielectric test | |
| Primary insulation (when tested in lieu of spark test) | 6.0 kilovolts (peak), 100 percent impulse test |
| Finished wire | 8.0 kilovolts (peak), or 3 kHz spark test 5.5 kV, 100 percent test |
| Insulation resistance | 5000 megohms for 1000 feet (min.) |
| Life cycle | Oven temperature, 200±2 °C (392±3.6 °F) for 120 hours |
| Physical properties of primary insulation | |
| Tensile strength | 2500 $lb/in^2$ (min.) |
| Elongation | 150 percent (min.) |
| Shrinkage | 0.125 inch (max.) at 250±2 °C (482±2 °F) |
| Spark test primary insulation | 1500 volts (root mean square), 3 kHz, 100 percent test |
| Surface resistance | 500 megohms inches (min.), initial and final readings |
| Thermal shock | Oven temperature, 150±2 °C (302±3.6 °F) |
| Maximum change in measurement | 0.060 inch |
| Wet dielectric test | 2500 volts (root mean square) |
| Wrap test | Mandrel wrap test required |



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INSPECTION:

| | |
|---|---|
| **Basic Electricals:** | |
| Conductor resistance (ohms/1000 feet at 25 °C, max.) | 46.01 |
| Voltage withstand (volts, root mean square, min.) | |
| Conductor to conductor and shield | 1000 |
| Component shield to component shield | Not applicable |
| Component shield to overall shield | Not applicable |
| Insulation resistance (megohms/1000 feet, min.) | |
| Conductor to conductor and shield | 2500 |
| Shield to shield (component shield only) | 100 |
| Shield to shield (component shield to overall shield) | 100 |
| Conductor and shield continuity | No failure |
| Jacket flaws | No failure |
| **Group A:** | |
| Visual and dimensional | No failure |
| Capacitance | |
| Mutual capacitance (at 1 MHz, picofarads/foot (pF/ft), max.) | 60 |
| Capacitance unbalance (percent, max.) | 15 |
| Characteristic impedance (at 1 MHz, ohms) | 70 (min.), 105 (max.) |
| Watertightness<br>(see table IX of MIL-DTL-24640 for limits of water leakage) | No failure |
| Weight | No failure |
| **Group B:** | |
| Cold bending | No failure |
| Drip | No failure |
| Cross-link proof test (percent, max.) | |
| Jacket | 50 |
| Tear (lb/in thickness, min.) | 35 |
| Physicals (unaged) | |
| Jacket | |
| Tensile strength ($lb/in^2$, min.) | 1300 |
| Elongation (percent, min.) | 160 |
| Attenuation (at 3 MHz, dB/100 feet, max.) | 3 |
| Gas flame (1 hour) | Not applicable |



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**Group C:**

| | |
|---|---|
| Physicals (aged), air oven | |
| Jacket | |
| Tensile strength (percent of unaged, min.) | 60 |
| Elongation (percent of unaged, min.) | 60 |
| Permanence of printing (conductor insulation, cycles, Method 1 only, min.) | 25 |
| Permanence of printing (jacket, cycles, min.) | 125 |
| Heat distortion (percent of unaged, max.) | 30 |
| Cable sealant removability | No failure |
| Shrinkage | No failure |
| Surface transfer impedance | |
| Milliohms per meter (max.) | 700 |
| EMP response (dB, min.) | 60 |

**Group D:**

| | |
|---|---|
| Flame propagation | No failure |

**Qualification Inspection:**

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

| | |
|---|---|
| Aging and compatibility (cable) | No failure |
| Acid gas equivalent (percent, max.) | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Halogen content (percent, max.) | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Immersion (jacket) | |
| Tensile strength (percent of unaged, min.) | 50 |
| Elongation (percent of unaged, min.) | 50 |
| Smoke index (max.) | |
| Jacket | 25 |
| Fillers | 45 |
| Insulation | 10 |



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| | |
|---|---|
| Toxicity index (max.) | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket) - Type A (hardness, min.) | 80 |
| Weathering (jacket) | No failure |
| Fungus resistance | No failure |
| Abrasion resistance | No failure |
| Gas flame (3 hour) | Not applicable |
| Component wire | No failure |

UNIT ORDERING LENGTHS:

| Type and size | Feet (nominal) | |
|---|---|---|
| 2XOW-6, 2XOW-18, 2XOW-24, 2XOW-42, 2XOW-60, 2XOW-77 | 6 & 18 | 500 |
| | 24 - 77 | 250 |

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-085)

Review Activities:
Army – AR, AV, CR, CR4
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.



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